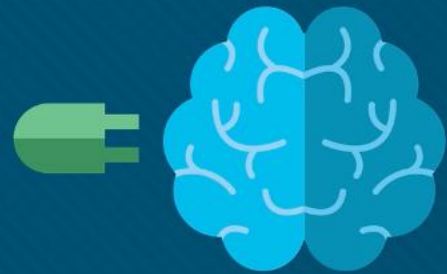




Module 10: Zone-Based Policy Firewalls

Networking Security v1.0
(NETSEC)



Module Objectives

Module Title: Zone-Based Policy Firewalls

Module Objective: Implement Zone-Based Policy Firewall using CLI.

Topic Title	Topic Objective
ZPF Overview	Explain how Zone-Based Firewalls are used to help secure a network.
ZPF Operation	Explain the operation of a Zone-Based Policy Firewall.
Configuring a ZPF	Configure a Zone-Based Firewall with CLI.

10.1 ZPF Overview

Benefits of a ZPF

There are two configuration models for Cisco IOS Firewall:

- **Classic Firewall** - The traditional configuration model in which firewall policy is applied on interfaces.
- **Zone-based Policy Firewall (ZPF)** - The new configuration mode in which interfaces are assigned to security zones, and firewall policy is applied to traffic moving between the zones.

There are several benefits of a ZPF:

- It is not dependent on ACLs.
- The router security posture is to block unless explicitly allowed.
- Policies are easy to read and troubleshoot with the Cisco Common Classification Policy Language (C3PL). C3PL is a structured method to create traffic policies based on events, conditions, and actions. This provides scalability because one policy affects any given traffic, instead of needing multiple ACLs and inspection actions for different types of traffic.
- Virtual and physical interfaces can be grouped into zones.
- Policies are applied to unidirectional traffic between zones.

ZPF Design

Designing ZPFs involves several steps:

Step 1. Determine the zones. A zone defines a boundary where traffic is subjected to policy restrictions as it crosses to another region of the network.

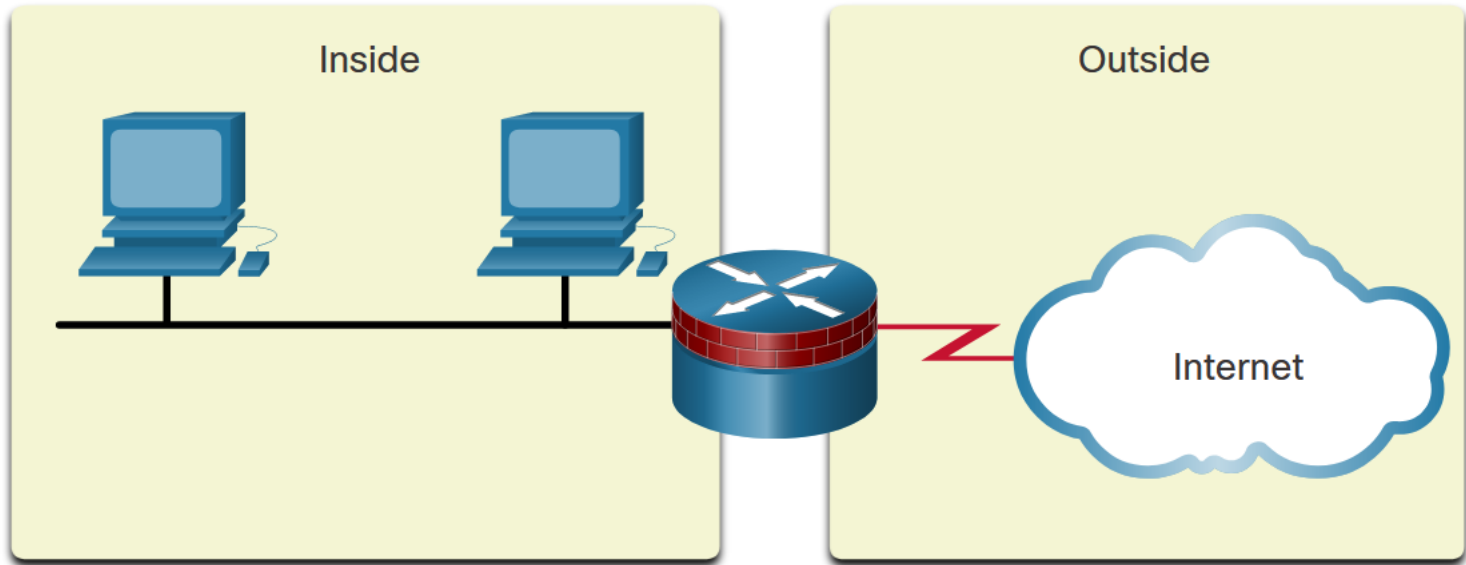
Step 2. Establish policies between zones. Define the sessions that clients in the source zones can request from servers in destination zones.

Step 3. Design the physical infrastructure. This includes dictating the number of devices between most-secure and least-secure zones and determining redundant devices.

Step 4. Identify subsets within zones and merge traffic requirements. Although an important consideration, implementing zone subsets is beyond the scope of this curriculum.

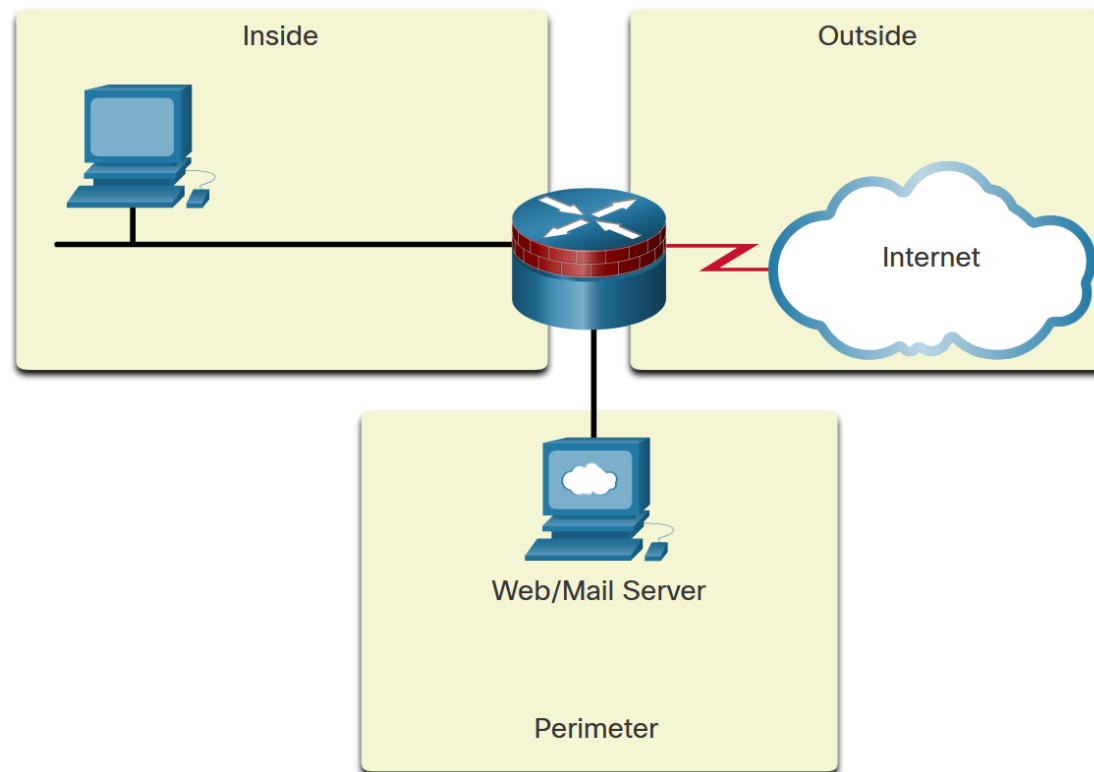
ZPF Design (Cont.)

LAN to Internet Example



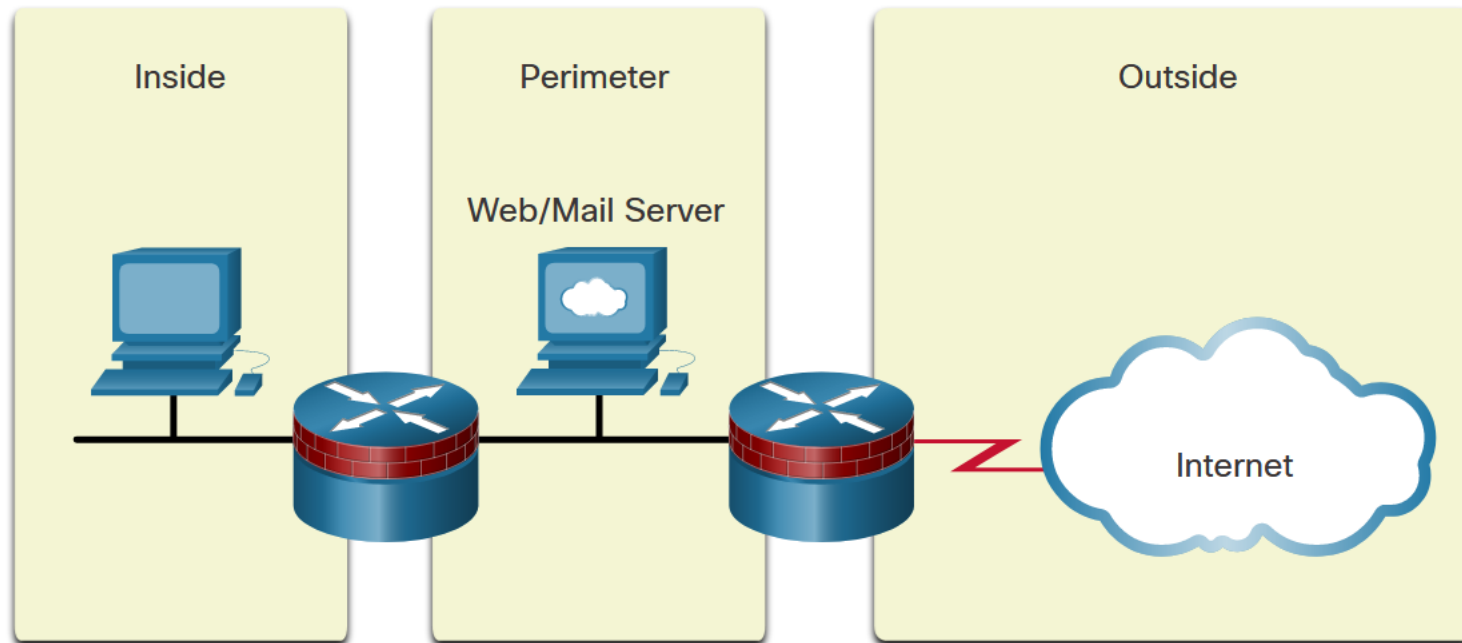
ZPF Design (Cont.)

Firewall with Public Servers Example 1



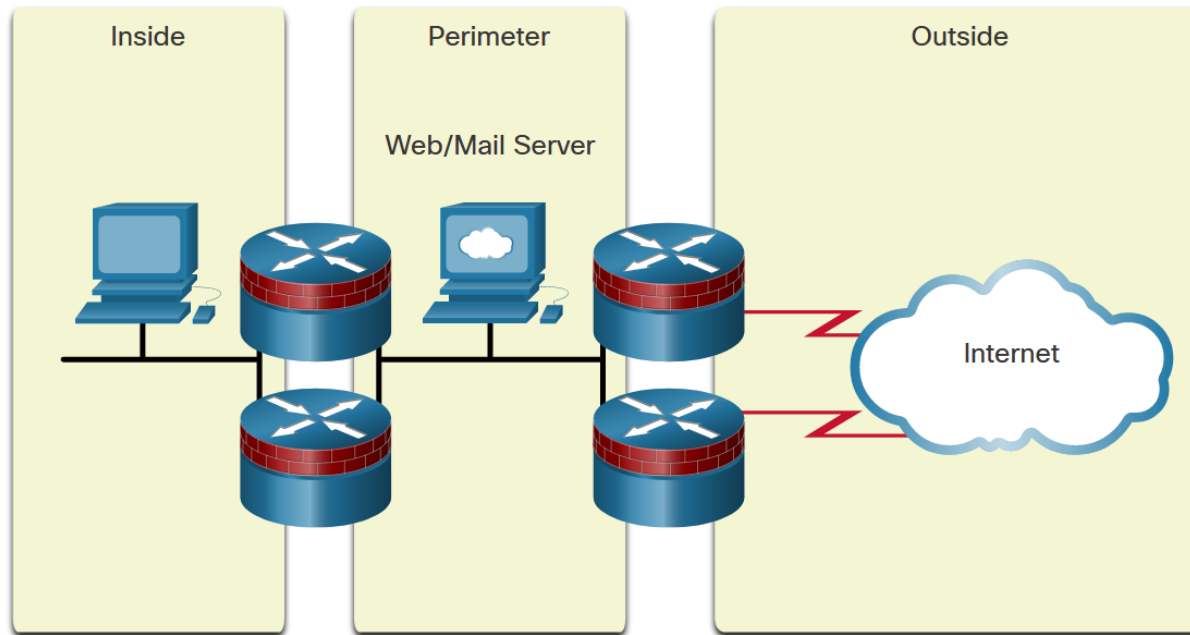
ZPF Design (Cont.)

Firewall with Public Servers Example 2



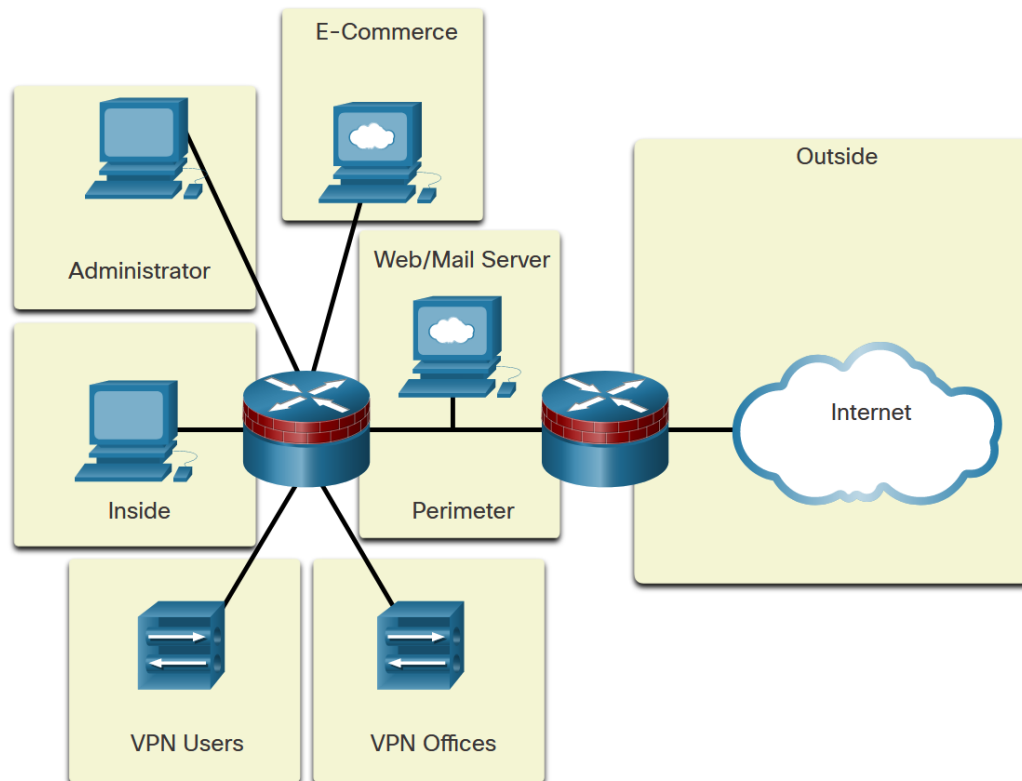
ZPF Design (Cont.)

Redundant Firewalls



ZPF Design (Cont.)

Complex Firewall



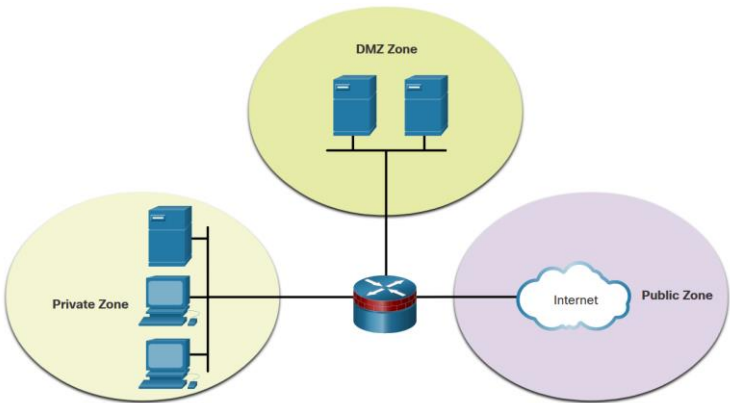
10.2 ZPF Operation

Policies identify actions that the ZPF will perform on network traffic. Three possible actions can be configured to process traffic by protocol, source and destination zones (zone pairs), and other criteria.

- **Inspect** - Performs Cisco IOS stateful packet inspection.
- **Drop** - Analogous to a **deny** statement in an ACL. A **log** option is available to log the rejected packets.
- **Pass** - Analogous to a **permit** statement in an ACL. The pass action does not track the state of connections or sessions within the traffic.

Rules for Transit Traffic

Traffic transiting through router interfaces is subject to several rules governing interface behavior. For the transit traffic example, refer to the topology shown in the figure.



Source Interface Member of Zone?	Destination Interface Member of Zone?	Zone-Pair Exists?	Policy Exists?	Result
NO	NO	N/A	N/A	PASS
YES	NO	N/A	N/A	DROP
NO	YES	N/A	N/A	DROP
YES (private)	YES (private)	N/A	N/A	PASS
YES (private)	YES (public)	NO	N/A	DROP
YES (private)	YES (public)	YES	NO	PASS
YES (private)	YES (public)	YES	YES	INSPECT

Rules for Traffic to the Self Zone

The self zone is the router itself and includes all the IP addresses assigned to the router interfaces. This is traffic that originates at the router or is addressed to a router interface.

The rules depend on whether the router is the source or the destination of the traffic, as shown in the table.

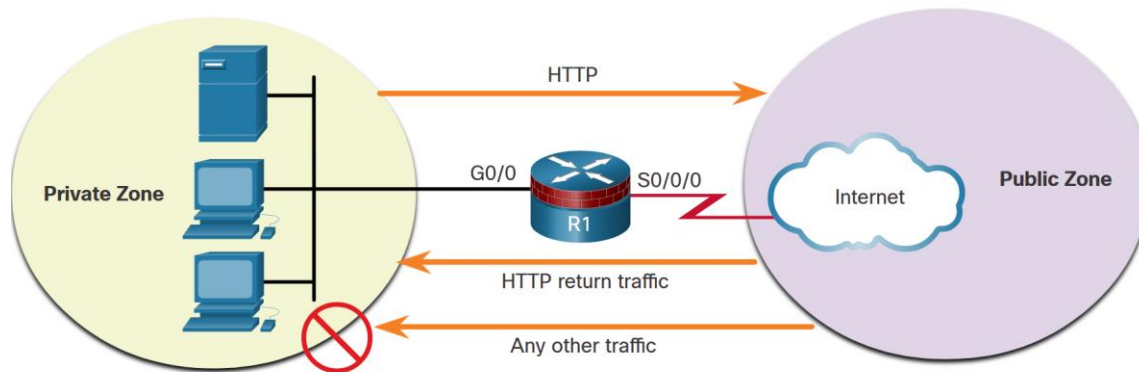
Source Interface Member of Zone?	Destination Interface Member of Zone?	Zone-Pair Exists?	Policy Exists?	Result
YES (self zone)	YES	NO	N/A	PASS
YES (self zone)	YES	YES	NO	PASS
YES (self zone)	YES	YES	YES	INSPECT
YES	YES (self zone)	NO	N/A	PASS
YES	YES (self zone)	YES	NO	PASS
YES	YES (self zone)	YES	YES	INSPECT

10.3 Configure a ZPF

Configure a ZPF

Configure a ZPF

The topology and steps shown in the figure will be used throughout the remainder of this topic to demonstrate ZPF configuration.



Step 1: Create the zones.

Step 2: Identify traffic with a class-map.

Step 3: Define an action with a policy-map.

Step 4: Identify a zone pair and match it to a policy-map.

Step 5: Assign zones to the appropriate interfaces.

Configure a ZPF

Step 1. Create Zones

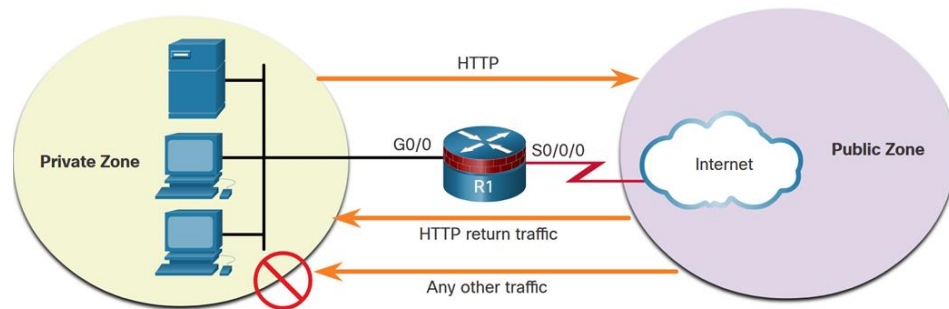
The first step, is to create the zones. However, before creating the zones answer a few questions:

- What interfaces should be included in the zones?
- What will be the name for each zone?
- What traffic is necessary between the zones and in which direction?

In the example topology, we have two interfaces, two zones, and traffic flowing in one direction. Traffic sourced from the public zone will not be allowed. Create the private and public zones for the firewall with the zone security command, as shown here.

```
Router(config)# zone security zone-name
```

```
R1(config)# zone security PRIVATE
R1(config-sec-zone)# exit
R1(config)# zone security PUBLIC
R1(config-sec-zone)# exit
R1(config)#
```



Step 2. Identify Traffic

The second step is to use a class-map to identify the traffic to which a policy will be applied.

The figure shows the syntax for the **class-map** command. There are several types of class-maps. For a ZPF configuration, use the **inspect** keyword to define a class-map. Determine how packets are evaluated when multiple match criteria exist. Packets must meet one of the match criteria (**match-any**) or all of the match criteria (**match-all**) to be considered a member of the class.

```
Router(config)# class-map type inspect [match-any | match-all] class-map-name
```

Parameter	Description
match-any	Packets must meet one of the match criteria to be considered a member of the class.
match-all	Packets must meet all of the match criteria to be considered a member of the class.
<i>class-map-name</i>	Name of the class-map that will be used to configure the policy for the class in the policy-map.

Step 2. Identify Traffic (Cont.)

The figure shows the syntax for the match statements in class-map sub-configuration mode. Match traffic to an ACL, a specific protocol, or even another class-map.

```
Router(config-cmap)# match access-group {acl-# | acl-name }  
Router(config-cmap)# match protocol protocol-name  
Router(config-cmap)# match class-map class-map-name
```

Parameter	Description
match access-group	Configures the match criteria for a class-map based on the specified ACL number or name.
match protocol	Configures the match criteria for a class-map based on the specified protocol.
match class-map	Uses another class-map to identify traffic.

Step 3. Define an Action

The third step is to use a policy-map to define what action should be taken for traffic that is a member of a class. The figure shows the command syntax to configure a policy-map. An action is a specific functionality. It is typically associated with a traffic class. For example, inspect, drop, and pass are actions.

```
R1(config)# policy-map type inspect policy-map-name  
R1(config-pmap)# class type inspect class-map-name  
R1(config-pmap-c)# {inspect | drop | pass}
```

Parameter	Description
inspect	An action that offers state-based traffic control. The router maintains session information for TCP and UDP and permits return traffic.
drop	Discards unwanted traffic
pass	A stateless action that allows the router to forward traffic from one zone to another

Step 4. Identify a Zone-Pair and Match to a Policy

The fourth step is to identify a zone pair and associate that zone pair to a policy-map. The figure shows the command syntax. Create a zone-pair with the **zone-pair security** command. Then use the **service-policy type inspect** command to attach a policy-map and its associated action to the zone-pair.

```
Router(config)# zone-pair security zone-pair-name source {source-zone-name |  
self} destination {destination-zone-name | self}  
Router(config-sec-zone-pair)# service-policy type inspect policy-map-name
```

Parameter	Description
source <i>source-zone-name</i>	Specifies the name of the zone from which traffic is originating.
destination <i>destination-zone-name</i>	Specifies the name of the zone to which traffic is destined.
self	Specifies the system-defined zone. Indicates whether traffic will be going to or from the router itself.

Step 5. Assign Zones to Interfaces

The fifth step is to assign zones to the appropriate interfaces. Associating a zone to an interface will immediately apply the service-policy that has been associated with the zone. If no service-policy is yet configured for the zone, all transit traffic will be dropped. Use the **zone-member security** command to assign a zone to an interface. In the example, GigabitEthernet 0/0 is assigned the PRIVATE zone, and Serial 0/0/0 is assigned the PUBLIC zone.

```
Router(config-if)# zone-member security zone-name
```

In the following example, GigabitEthernet 0/0 is assigned the PRIVATE zone, and Serial 0/0/0 is assigned the PUBLIC zone.

```
R1(config)# interface GigabitEthernet 0/0
R1(config-if)# zone-member security PRIVATE
R1(config-if)# interface Serial 0/0/0
R1(config-if)# zone-member security PUBLIC
```

Configure a ZPF

Verify a ZPF Configuration

Verify a ZPF configuration by viewing the running configuration. Notice that the class-map is listed first. Then the policy-map makes use of the class-map.

The zone configurations follow the policy-map configurations with zone naming, zone pairing, and associating a service-policy to the zone pair.

Finally, the interfaces are assigned zones.

```
R1# show run | begin class-map
!
<some output omitted>
!
class-map type inspect match-any HTTP-TRAFFIC
  match protocol http
  match protocol https
  match protocol dns
!
policy-map type inspect PRIV-TO-PUB-POLICY
  class type inspect HTTP-TRAFFIC
    inspect
  class class-default
    drop
!
zone security PRIVATE
zone security PUBLIC
zone-pair security PRIV-PUB source PRIVATE destination PUBLIC
  service-policy type inspect PRIV-TO-PUB-POLICY
!
interface GigabitEthernet0/0
  zone-member security PRIVATE
!
interface Serial0/0/0
  zone-member security PUBLIC
!
```

Configure a ZPF

Verify a ZPF Configuration (Cont.)

The example shows verification information after a test of the ZPF configuration. A PRIVATE zone host 192.168.1.3 established an HTTPS session with a web server at 10.1.1.2. Notice further down in the command output that four packets matched the **class class-default**. This verification information was generated by having host 192.168.1.3 ping the web server at 10.1.1.2.

Other commands include:

- **show class-map type inspect**
- **show zone security**
- **show zone-pair security**
- **show policy-map type inspect**



```
R1# show policy-map type inspect zone-pair sessions

policy exists on zp PRIV-PUB
Zone-pair: PRIV-PUB

Service-policy inspect : PRIV-TO-PUB-POLICY

Class-map: HTTP-TRAFFIC (match-any)
  Match: protocol http
    12 packets, 384 bytes
    30 second rate 0 bps
  Match: protocol https
    5 packets, 160 bytes
    30 second rate 0 bps
  Match: protocol dns
    0 packets, 0 bytes
    30 second rate 0 bps

Inspect

Number of Established Sessions = 1
Established Sessions
  Session 2204E220 (192.168.1.3:1049)=>(10.1.1.2:443)
https:tcp
  SIS_OPEN/TCP_CLOSEWAIT
    Created 00:00:14, Last heard 00:00:11
    Bytes sent (initiator:responder) [821:1431]

Class-map: class-default (match-any)
  Match: any
  Drop
    4 packets, 160 bytes

R1#
```


ZPF Configuration Considerations

When configuring a ZPF with the CLI, there are several factors to consider:

- The router never filters the traffic between interfaces in the same zone.
- An interface cannot belong to multiple zones. To create a union of security zones, specify a new zone and appropriate policy map and zone pairs.
- ZPF can coexist with Classic Firewall although they cannot be used on the same interface. Remove the **ip inspect** interface configuration command before applying the **zone-member security** command.
- Traffic can never flow between an interface assigned to a zone and an interface without a zone assignment. Applying the **zone-member** configuration command always results in a temporary interruption of service until the other zone-member is configured.
- The default inter-zone policy is to drop all traffic unless otherwise specifically allowed by the service-policy configured for the zone-pair.
- The **zone-member** command does not protect the router itself (traffic to and from the router is not affected) unless the zone- pairs are configured using the predefined self zone.

